ST+CoE figure-flow draft: homology to decreasing ρ to Migkas $A(\ell)$ fits to sky rings

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draft

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Abstract

This draft follows the figure order of nine Telegram screenshots (2026-09-22 ∼11:30 CDT). Pure homologous Sedov–Taylor (ST) expansion about a real center of expansion (CoE) yields an isotropic apparent Hubble law for every observer. Anisotropy enters only through Gauss gravity on a radially decreasing density. Against the Migkas et al. (2020) $A(\ell)$ strip, first-principles ST+Gauss geometry alone is modest; Amp$\times$ sin phase rotation and a multiplicative 2ℓ residual layer raise R_A^2 substantially but remain fit templates until derived from time-integrated gravity. Bare Gauss + decreasing-ρ leading multipole is a quadrupole, not a pure dipole. The CMB is parked as an open adversary. Numbers below are taken from on-disk JSON/CSV/npz sidecars; missing metrics are marked TBD. Two figure slots at the end are empty for later additions.

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1 Locked parameters (on-disk)

Table 1: Locked numbers from sidecars under `figures/`, `map_compare/amp_mean_param_noise/`, and `NOTES_FOR_CHRIS.md`. No invention.

Quantity	Value (source)
Homology	$\vec{v} = H\vec{R}$; $v_{\rm rel} = H\ell$
Toy density	$\rho(r)/\rho_0 = \mathrm{e}^{-r/L}$; $D/L = 3$ (journal fig)
$\kappa_{\rm natural}$	$\Omega_m/2 = 0.15$ for $\Omega_m = 0.3$
Axisym. before / after sine-analogue	$R^2 \approx 0.307 \to 0.553$; RMSE $0.059 \to 0.047$
Amp$\times$ sin (A scan peak)	$\mathrm{Amp}_A = -0.06$, $\phi \approx 11.46°$
Shared-opp. Amp_B	$\mathrm{Amp}_B = +0.06$ at same ϕ; $R^2_A \approx 0.746$, $R^2_B \approx 0.790$
2ℓ A-tuned	$\varepsilon \approx -0.043$, $\psi \approx 176°$; $R^2_A \approx 0.875$; RMSE $0.038 \to 0.029$
2ℓ A-tuned R^2_B after	TBD (CSV stores pre-2ℓ R^2_B only)
2ℓ B-tuned	$\varepsilon \approx +0.006$, $\psi \approx 38°$; $R^2_B \approx 0.790$, $R^2_A \approx 0.766$
After-2ℓ strip extrema	MAX $\approx 30°, 185°$; MIN $\approx 115°, 265°$
Highlight bands	$18°$ end-to-end $= 5\%$ of $360° = \pm 9°$
Pose (this draft)	anti-CoE $(\ell, b) = (0, 0)$; CoE $(180, 0)$
Migkas A map extrema	max $(17°, -9°)$; min $(274°, -22°)$
Migkas dipole center	$(263°, -21°)$ at 3.22σ (published)
CMB schematic	vision shell radius r; $S_n \to S_{n+1}$ spacing $2r$

2 Homologous blast: isotropic H for every observer

Place a real geometric CoE at the origin. Label the observer O at $\vec{R}_O$ and a probe P at $\vec{R}_P$. Homologous expansion means every tracer moves radially from the CoE with the same H:

$$\vec{v}(\vec{R}) = H\,\vec{R}. \tag{1}$$

Absolute velocities of P and O are $\vec{v}_P = H\vec{R}_P$ and $\vec{v}_O = H\vec{R}_O$. The relative velocity is therefore

$$\vec{v}_{\rm rel} = \vec{v}_P - \vec{v}_O = H\big(\vec{R}_P - \vec{R}_O\big) = H\,\vec{\ell}, \tag{2}$$

where $\vec{\ell} = \vec{R}_P - \vec{R}_O$ is the observer–probe separation. The line-of-sight recession speed is $v_{\rm los} = \vec{v}_{\rm rel} \cdot \hat{\ell} = H\,\ell$, so

$$H_{\rm obs} \equiv \frac{v_{\rm los}}{\ell} = H. \tag{3}$$

The CoE drops out of the relative-velocity measurement. Every off-center observer measuring relative velocities sees the same isotropic H. Being off-center under pure homology does *not* produce a sky dipole in $H_{\rm obs}$. Something must break homology—here, gravity.

3 Decreasing density, enclosed mass, and Gauss g

A uniform ball gives $\vec{g} \propto -\vec{r}$ inside, which preserves homologous flow and cannot source a sky anisotropy in $H_{\rm obs}$. Reject that. Take a spherical, radially decreasing density (toy exponential used in the journal figure)

$$\frac{\rho(r)}{\rho_0} = \mathrm{e}^{-r/L}, \tag{4}$$

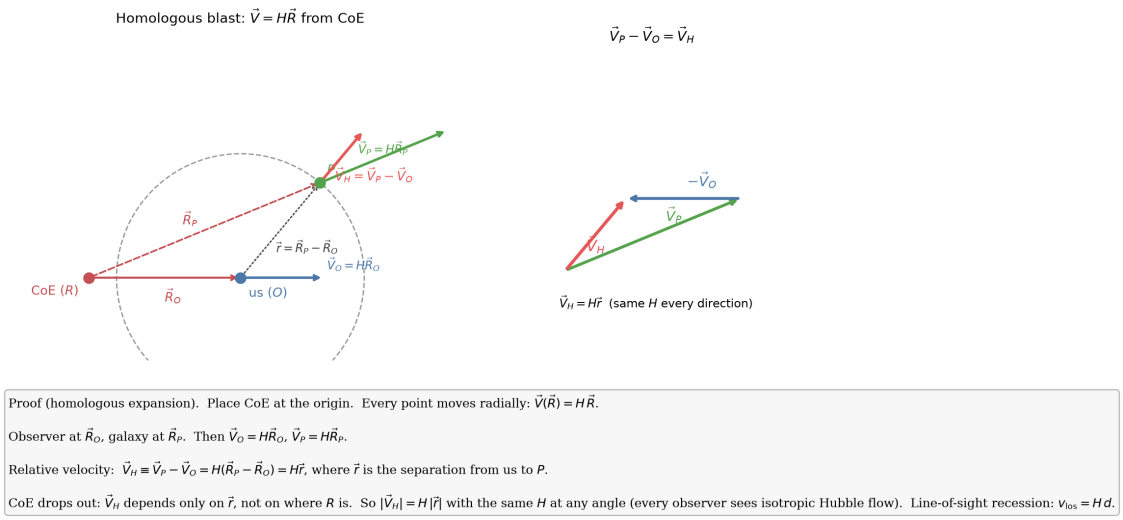


Figure 1: Homologous blast: absolute velocities point away from the CoE, but $\vec{v}_{\rm rel} = H\vec{\ell}$ so every observer measures isotropic $H_{\rm obs} = H$.

with scale length L. The enclosed-mass fraction with $X \equiv r/L$ is

$$\frac{m(r)}{M} = 1 - \mathrm{e}^{-X}\left(1 + X + \tfrac{1}{2}X^2\right), \tag{5}$$

which follows by integrating $4\pi r^2\rho(r)$ and normalizing to the total mass M of the infinite exponential. Gauss / shell theorem then gives

$$\vec{g}(r) = -\frac{G\,M(<r)}{r^2}\,\hat{r}. \tag{6}$$

At the observer distance D, a natural dimensionless gravity strength is

$$\kappa_{\rm natural} = \frac{G\,M_{\rm enc}(<D)}{D^3H^2} = \frac{\Omega_m}{2} = 0.15 \quad \text{for } \Omega_m = 0.3. \tag{7}$$

The journal toy panel draws the observer at $D/L = 3$ (not the later recommended-profile $D/L \approx 5.9$). Leading multipole of off-center Gauss+decreasing-ρ is a quadrupole (P_2), not a pure dipole—keep that honest when reading later Amp$\times$ sin / 2ℓ templates.

4 Ideal-integration ST+Gauss: A vs angle (sheet AF)

Geometry: CoE at the origin, observer at distance D, sources sampled at observer-centric depth d, Earth-centric angle measured from the preferred axis $\hat{D}$. The Migkas A proxy used on the sheet AF series is

$$A \;\propto\; \frac{H_{\rm obs}}{H} \qquad \text{(sheet AF; mean-renormed on the strip as needed).} \tag{8}$$

(Alternate cards use an inverse-square luminosity-distance mapping $A_{\rm raw} \propto 1/(H_{\rm obs}/H)^2$; the ideal-integration panel below is the sheet AF H-ratio series.) First-principles ST+Gauss geometry alone (no Amp$\times$ sin) reaches only $R_A^2 \approx 0.543$ on the digitized Migkas Fig. 6 strip (summary CSV: `geometry_or_AN_alone`).

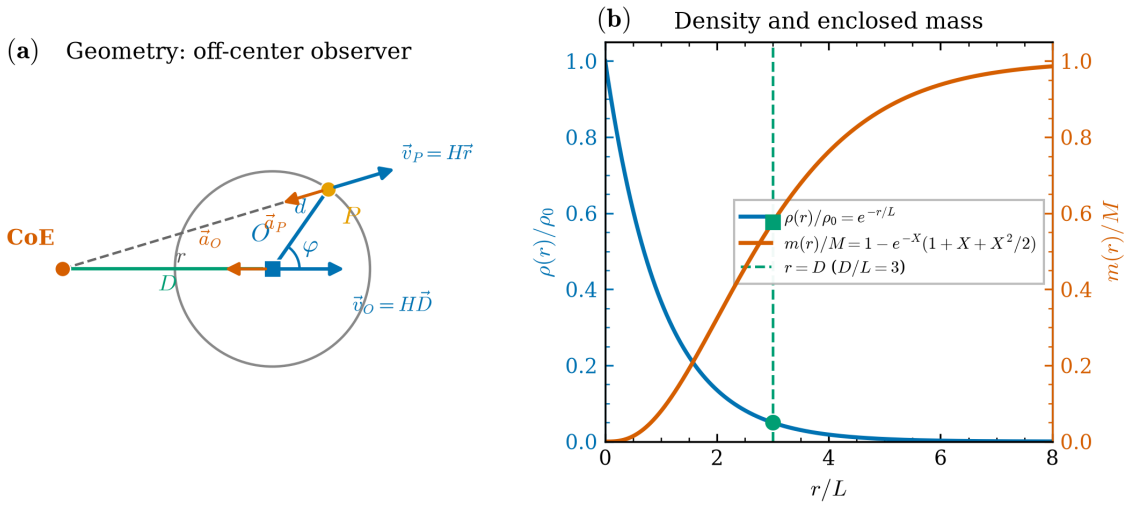


Figure 2: Toy exponential $\rho(r)/\rho_0 = \mathrm{e}^{-r/L}$ and enclosed-mass fraction $m(r)/M$ from Eq. (5). Observer mark at $D/L = 3$ as in the journal figure sidecar.

5 Axisymmetric AF + sine-rotation analogue + residuals

Before the Amp× sin template, an axisymmetric rotation analogue compares $\mathrm{AF}_{\mathrm{smooth}}$ to a rotated smooth curve. On-disk sidecar `A_vs_angle_axisymmetric_before_rotation_with_residuals.json`:

$$R^2_{\mathrm{before}} \approx 0.307, \quad \mathrm{RMSE}_{\mathrm{before}} \approx 0.059; \qquad R^2_{\mathrm{after}} \approx 0.553, \quad \mathrm{RMSE}_{\mathrm{after}} \approx 0.047. \tag{9}$$

This panel is *not* Amp× sin and *not* 2ℓ; it is the axisymmetric sine-rotation analogue only.

6 Amp× sin + multiplicative 2ℓ correction

6.1 Amp× sin convention (verified)

Sheet / level-set / cone-scan convention (not first-principles Gauss):

$$A'(\ell) = \big(1 - \mathrm{Amp}_A \cdot \sin(\Omega\ell + \phi)\big)\, A_{\mathrm{physics}}, \qquad \Omega = \frac{2\pi}{360}, \tag{10}$$

with ℓ in degrees and ϕ in radians in the same trig argument (sidecar scripts use ϕ in rad). Associated B uses the same sine with an independent amplitude Amp_B (shared-opp scans often take $\mathrm{Amp}_B = -\mathrm{Amp}_A$). Summary CSV peak for A: $\mathrm{Amp}_A = -0.06$, $\phi \approx 11.46°$, $R^2_A \approx 0.746$. Shared-opp at that peak: $\mathrm{Amp}_B = +0.06$, $R^2_B \approx 0.790$.

6.2 2ℓ multiplier

Multiplicative residual layer tuned on the Amp× sin A curve (`best_fit_2ell_multiplier.npz`):

$$A_{\mathrm{after}}(\ell) = \big(1 + \varepsilon \cdot \sin(2\ell_{\mathrm{rad}} + \psi)\big)\, A_{\mathrm{before}}(\ell), \tag{11}$$

with $\ell_{\mathrm{rad}} = \ell° \cdot \pi/180$, $\varepsilon \approx -0.043$, $\psi \approx 176°$ (A-tuned). On disk: $R^2_A \approx 0.746 \to 0.875$; RMSE $\approx 0.038 \to 0.029$.

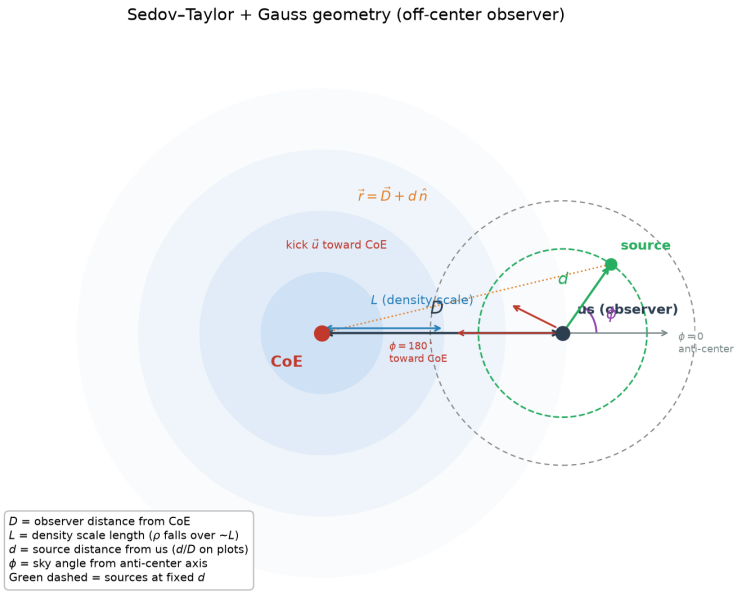


Figure 3: Optional prior geometry: CoE, observer at D, source depth d, angle from the preferred axis.

After-2ℓ strip extrema (from $+x$ / anti-CoE in the pose of Sec. 7): MAX $\approx 30°$ and $185°$; MIN $\approx 115°$ and $265°$. Highlight bands are

$$5\% \text{ of } 360° = 18° \text{ end-to-end} = \pm 9° \text{ about each extremum.} \tag{12}$$

7 Three-panel CoE rings (2 MAX + 2 MIN)

Pose for this draft: anti-CoE at $(\ell, b) = (0°, 0°)$, CoE at $(180°, 0°)$; Mollweide with ℓ increasing LEFT ($\text{lon}_{\text{disp}} = -\ell_{\text{wrapped}}$). Rings are small circles of constant $\mu = \hat{n} \cdot \hat{x}$ at the four after-2ℓ strip stops, each drawn as one $18°$ end-to-end highlight ($\pm 9°$) plus a mid-stroke.

Real Migkas A-map extrema: max $(17°, -9°)$, min $(274°, -22°)$. Honest note: $(185°, 0°)$ and $(115°, 0°)$ were strip stand-ins if still drawn on the three-panel figure.

8 Migkas Fig. 8 $\theta = 75°$ overlay

Migkas et al. (2020) Fig. 8 ($\theta = 75°$ cone A map): lowest A near $(274°, -22°)$, highest A near $(17°, -9°)$; published dipole center $(263°, -21°)$ at 3.22σ. Overlay: same four CoE rings / bands as Sec. 7. B has no published 2D map extrema (slope flat; fixed to Ball in the Fig. 6 strip analysis).

9 Amp/ϕ level-set families and R^2 heatmaps

Amp$\times$ sin is scanned in (Amp, ϕ) for A and B. Level-set families and heatmaps below are diagnostics: they map template depth/phase, not a derived multipole from Gauss.

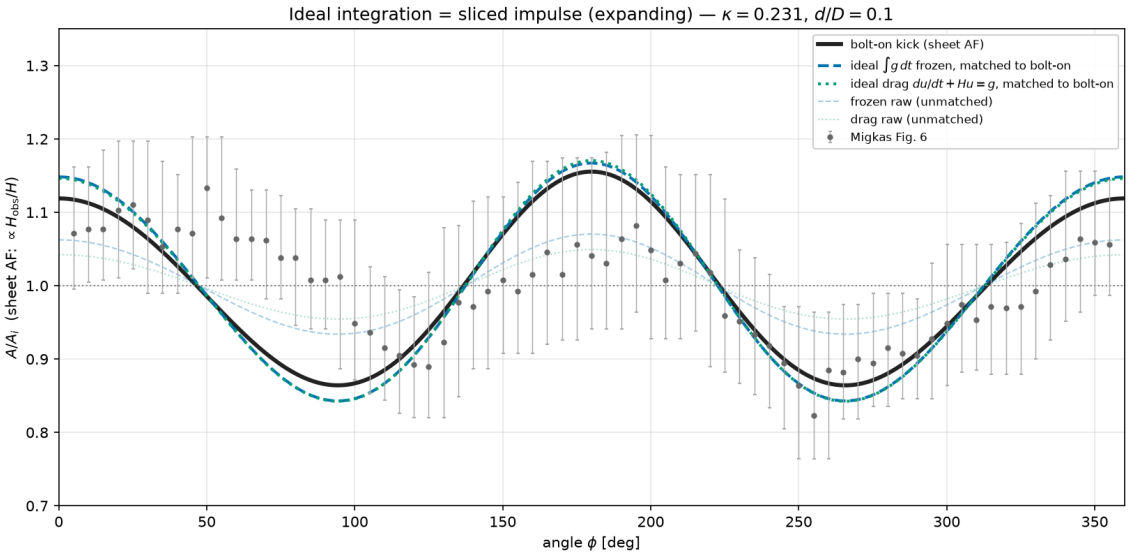


Figure 4: Ideal-integration ST+Gauss A vs Galactic longitude (sheet AF), before Amp$\times$ sin or 2ℓ.

10 Best A+associated B / best B+associated A with 2ℓ

From `R2_best_A_B_Amp_phi_summary.csv` and the 2ℓ npz files:

A-tuned. $\mathrm{Amp}_A = -0.06$, $\phi \approx 11.5°$, $\mathrm{Amp}_B = +0.06$; 2ℓ: $\varepsilon \approx -0.043$, $\psi \approx 176°$; $R^2_A \approx 0.875$ (was 0.746). Pre-2ℓ shared-opp $R^2_B \approx 0.790$; post-2ℓ associated R^2_B is TBD on the summary CSV (screenshot flow quoted ~ 0.663).

B-tuned. $\mathrm{Amp}_B = -0.06$, $\phi \approx 184.9°$, $\mathrm{Amp}_A = +0.06$; 2ℓ: $\varepsilon \approx +0.006$, $\psi \approx 38°$; $R^2_B \approx 0.790$, $R^2_A \approx 0.766$ (`best_fit_2ell_multiplier_from_B.npz`).

11 Amp$\times$ sin diagnostic Mollweide family

Full-sky Amp$\times$ sin diagnostic cards (physics baseline fixed; only Amp and ϕ change). Not first-principles Gauss.

12 Patchwork draws

Patchwork source map and four random draws with all markers. Fallback filenames (`map_09_patchwork_random.png` `GRID_patchwork_4draws_OWN_EXTREMA.png` / `_EXTREMA.png`) exist on disk but are not needed—preferred files are present.

13 CMB / CoE shell schematic (deferred adversary)

Park the CMB as an open adversary. The schematic geometry: vision shell radius $= r$; concentric shell spacing $S_n \to S_{n+1} = 2r$. A blackbody-shell dipole budget is a separate quantitative plot; this figure is only the geometric cartoon. No closure claimed here.

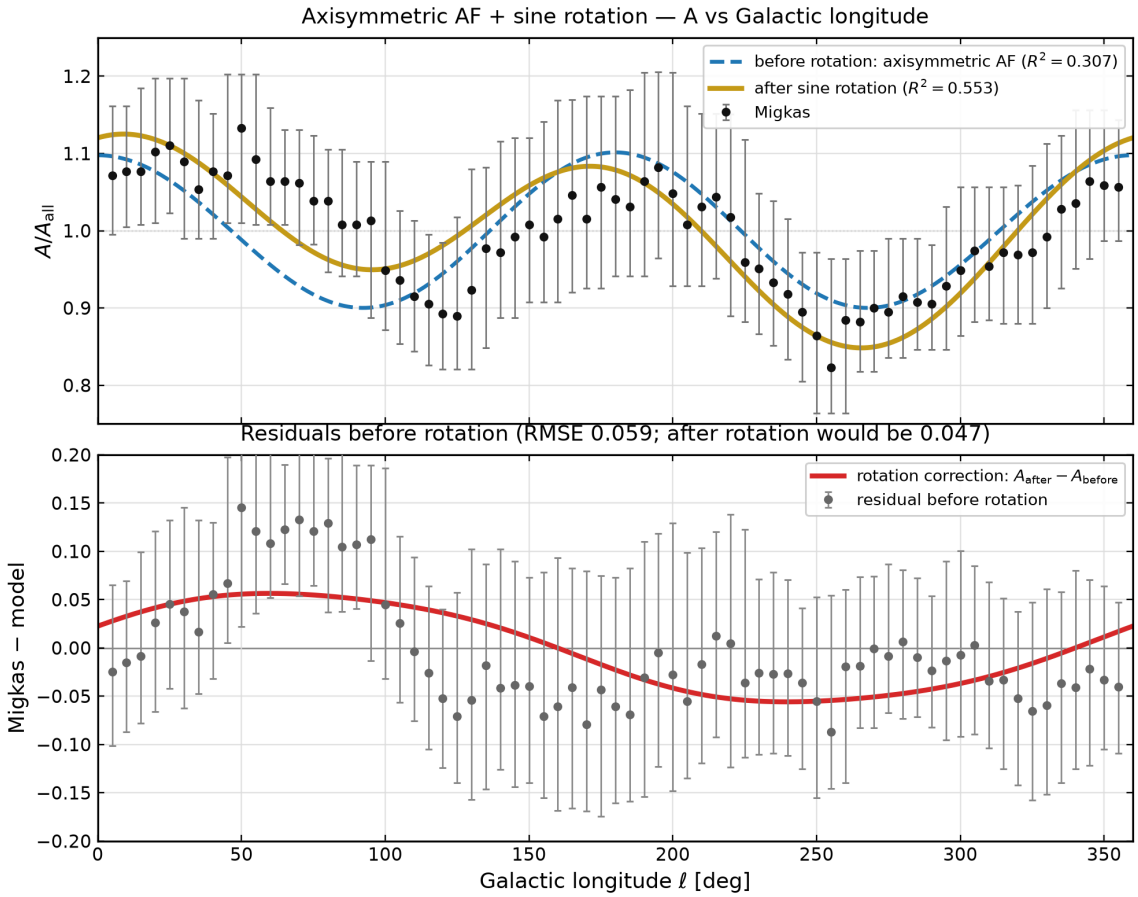


Figure 5: Axisymmetric AF before/after sine-rotation analogue, with residuals. $R^2 \approx 0.307 \rightarrow 0.553$; RMSE $\approx 0.059 \rightarrow 0.047$.

14 Figures to add

Acknowledgments

Wolfgang Rindler; Luke Ackerman (genesis).

References

[1] K. Migkas, G. Schellenberger, T. H. Reiprich, F. Pacaud, M. E. Ramos-Ceja, and L. Lovisari, *Probing cosmic isotropy with a new X-ray galaxy cluster sample through the L_X–T relation*, arXiv:2004.03305 (2020).

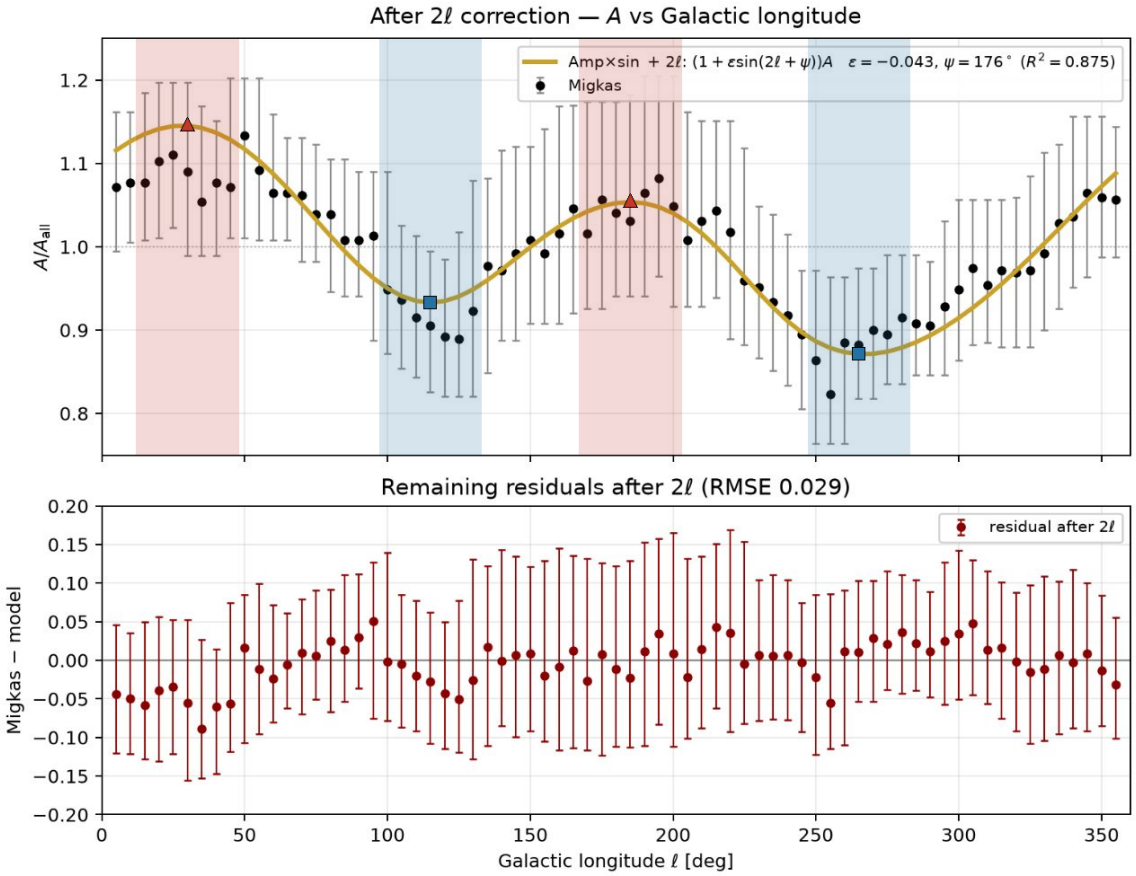


Figure 6: Amp× sin $A(\ell)$ after the 2ℓ multiplier, with residuals. Extrema near 30°/185° (MAX) and 115°/265° (MIN); bands ±9° (18° end-to-end). Fit template—not yet derived from time-integrated gravity.

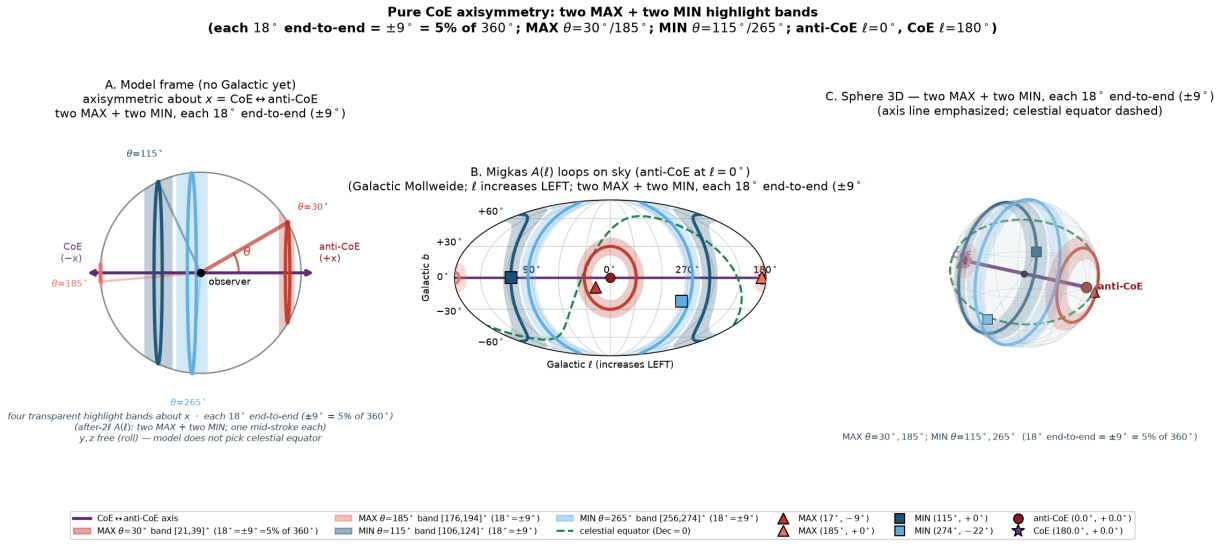


Figure 7: Three-panel CoE rings: strip stops, Mollweide, and sphere. Pose anti-CoE $(0, 0)$ / CoE $(180, 0)$. Real Migkas extrema $(17, -9)$ / $(274, -22)$; strip stand-ins $(185, 0)$ / $(115, 0)$ if present.

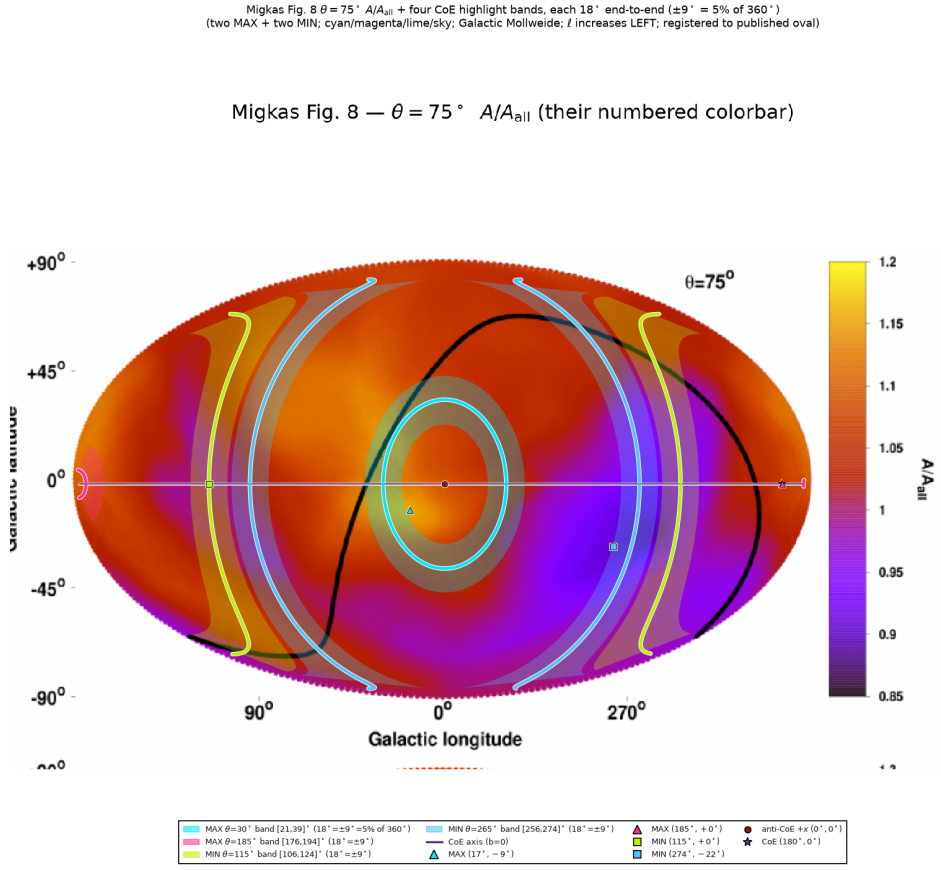
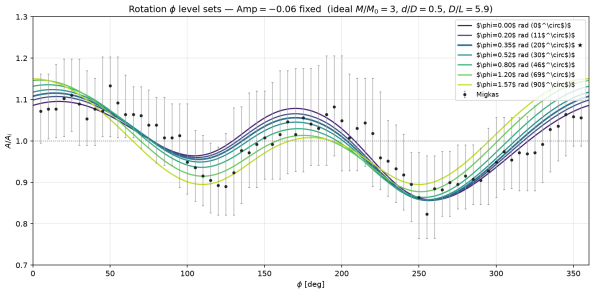
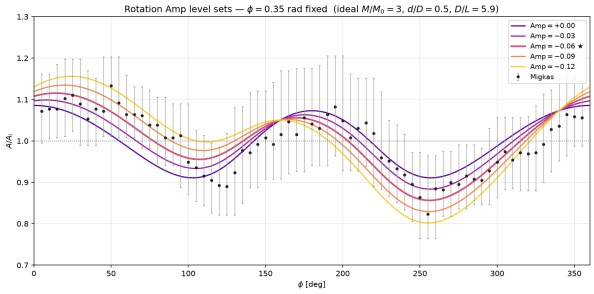


Figure 8: Migkas Fig. 8 $\theta = 75°$ with CoE-ring overlay. Max A $(17, -9)$, min A $(274, -22)$; dipole center $(263, -21)$ at 3.22σ.

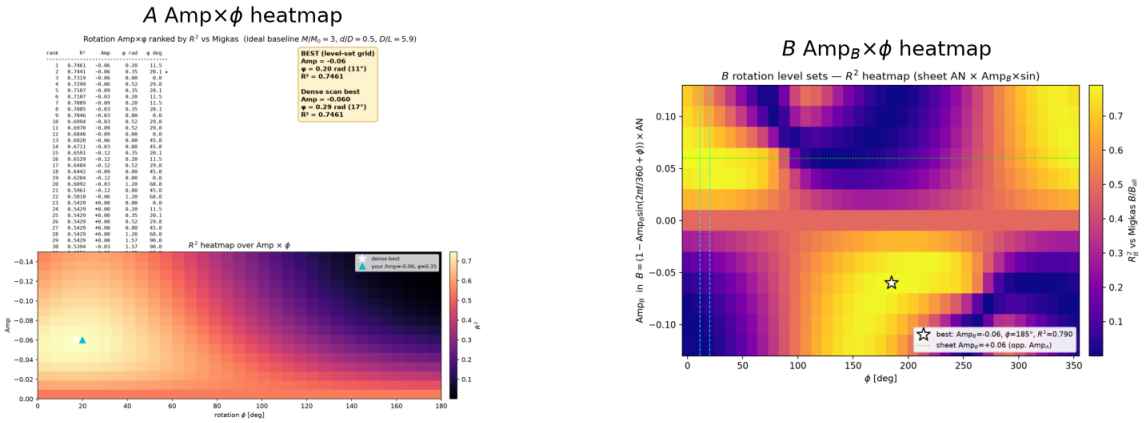


(a) ϕ level sets.



(b) Amp level sets.

Figure 9: Rotation level-set families in ϕ and Amp.



Best R^2 — Amp×sin rotation level sets (sheet AF / AN)

Model	R^2_A	R^2_B	Parameters
Geometry / AN alone (no Amp×sin)	0.543	0.483	—
Best Amp×φ (solo scan)	0.746	0.790	A: Amp=-0.06, φ=11.5° B: Amp_B=-0.06, φ=184.9°
Shared-opp at A best φ (Amp_B=−Amp_A)	0.746	0.790	Amp_A=-0.06, Amp_B=+0.06, φ≈11.5°
A + multiplicative 2ℓ (A-tuned)	0.875	— (hurts B if re	ε≈−0.043, ψ≈176°

Figure 10: R^2 table and heatmaps for best A/B Amp×ϕ scans (summary CSV companion).

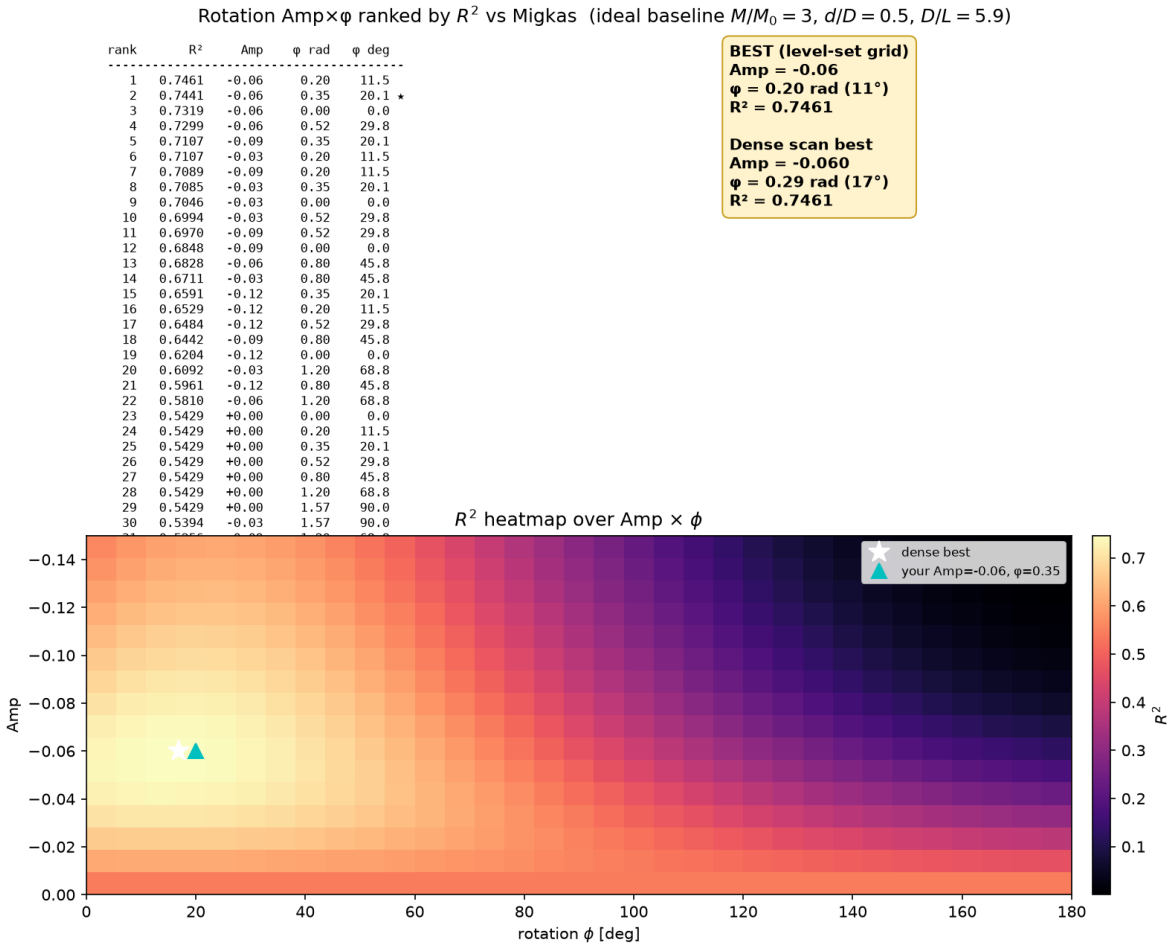


Figure 11: Ranked Amp×ϕ R^2 curves.

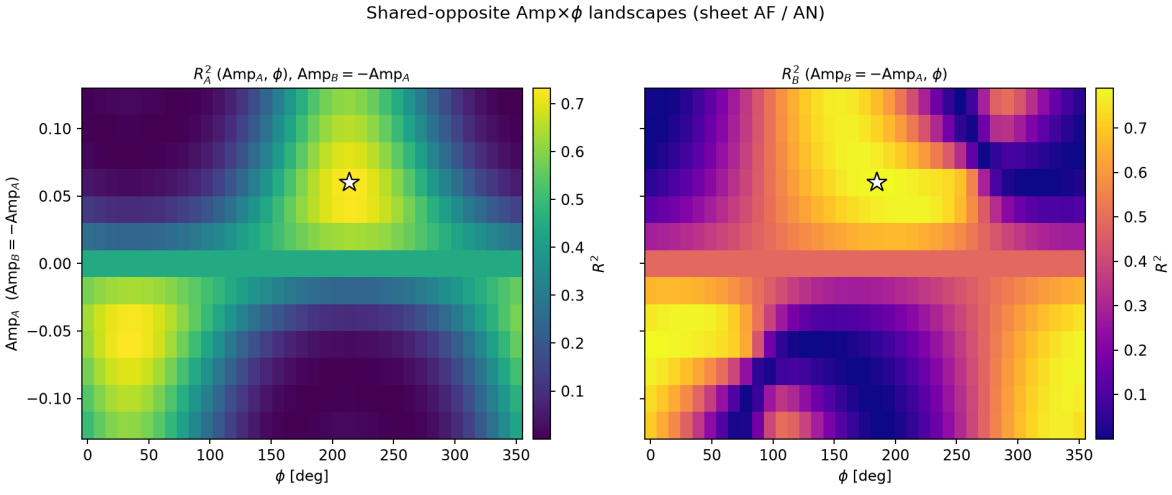


Figure 12: Shared $\text{Amp}_A/\text{Amp}_B$ heatmaps (present on disk).

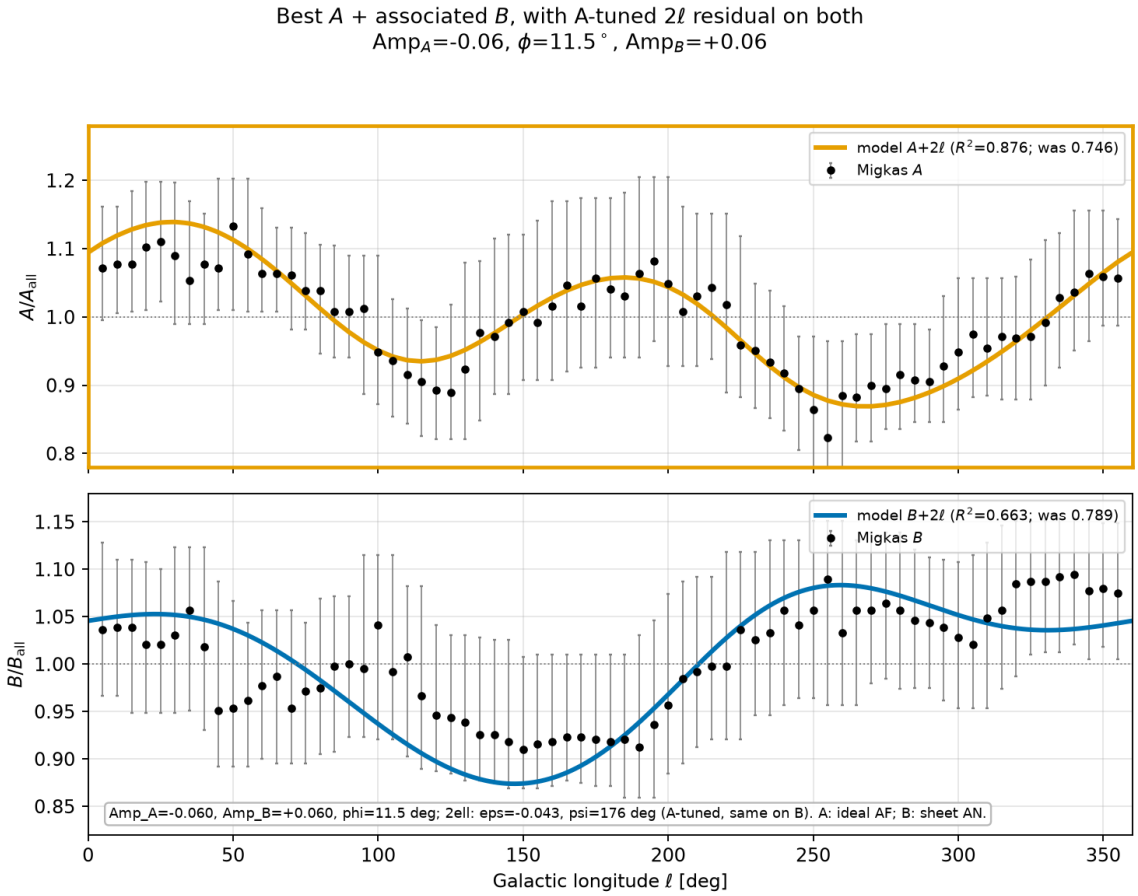


Figure 13: Best A (Amp$\times$ sin+2ℓ) with associated B.

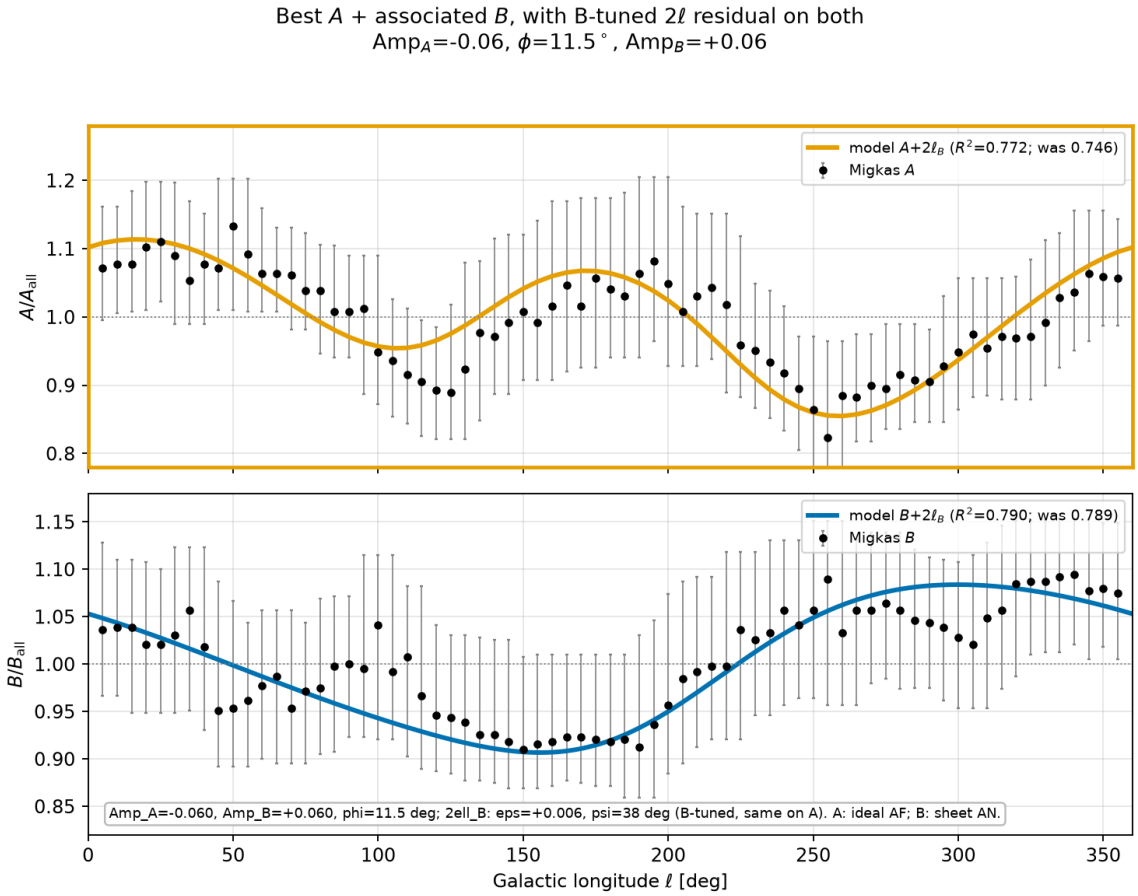


Figure 14: Best B-tuned Amp$\times$ sin+2ℓ with associated A (`_fromB` panel).

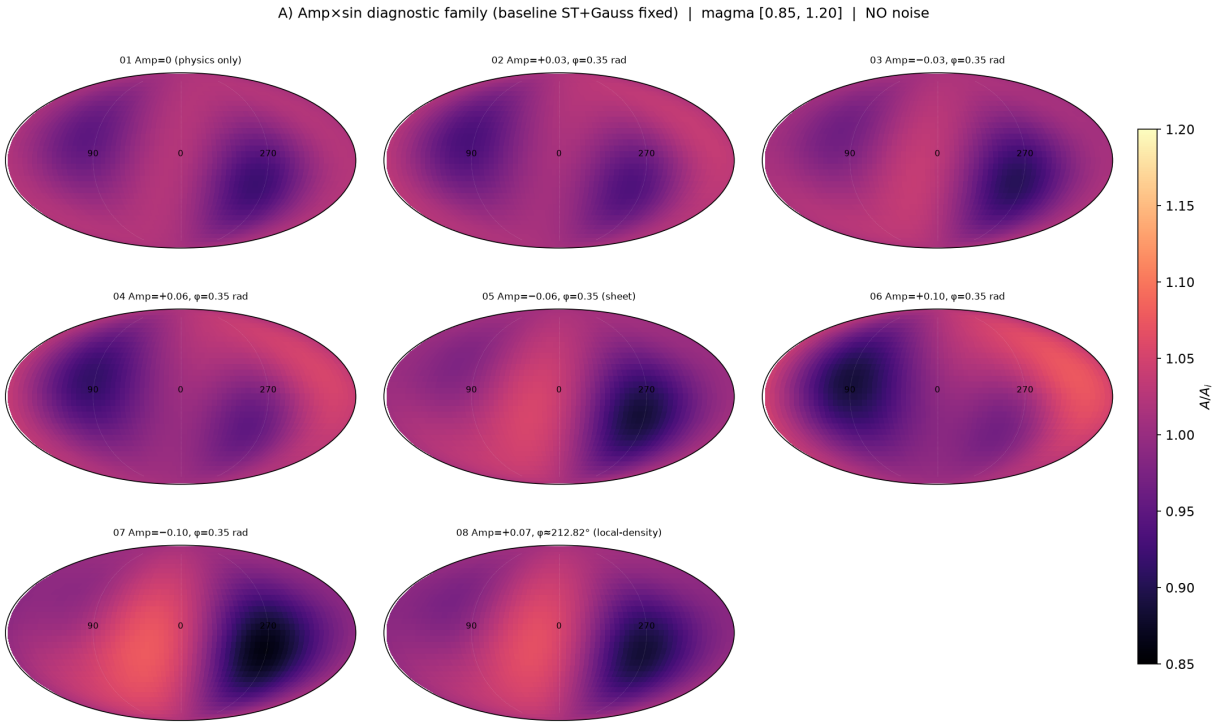


Figure 15: Amp× sin diagnostic Mollweide family grid.

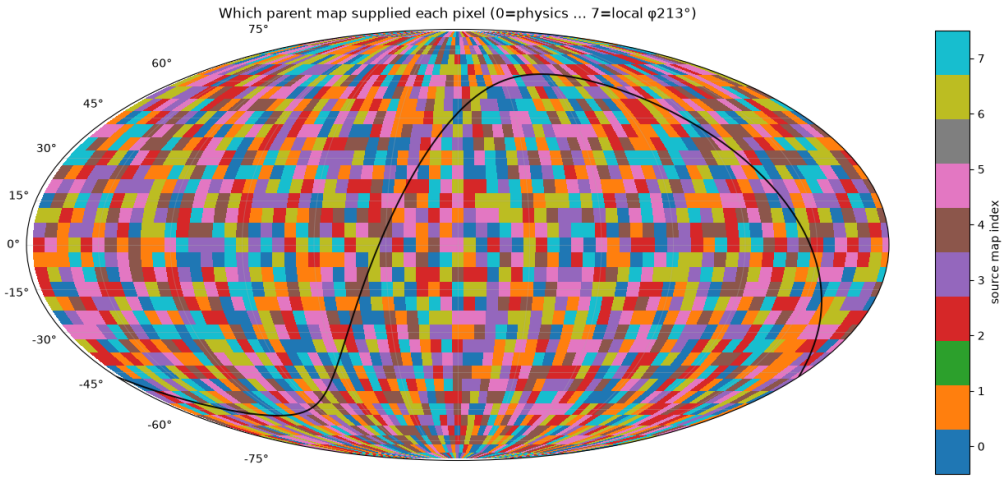


Figure 16: Patchwork sources map.

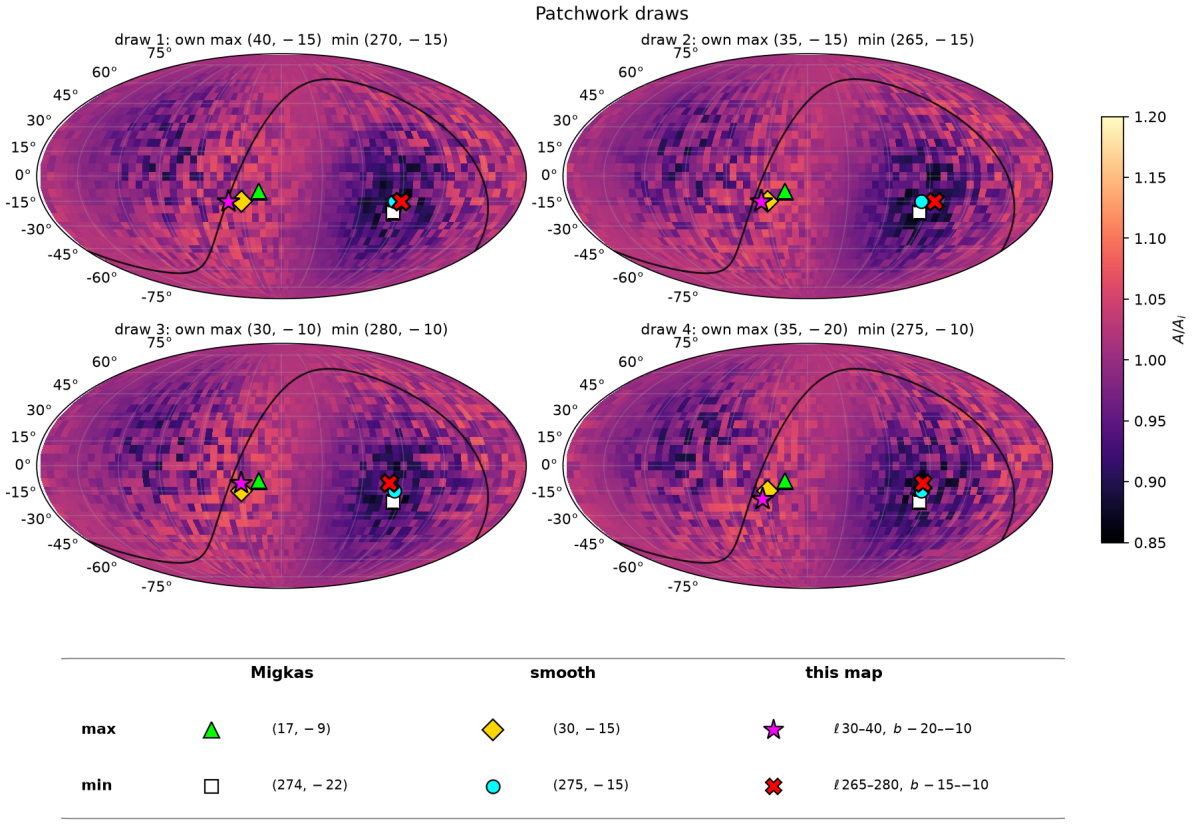


Figure 17: Four patchwork draws with all markers.

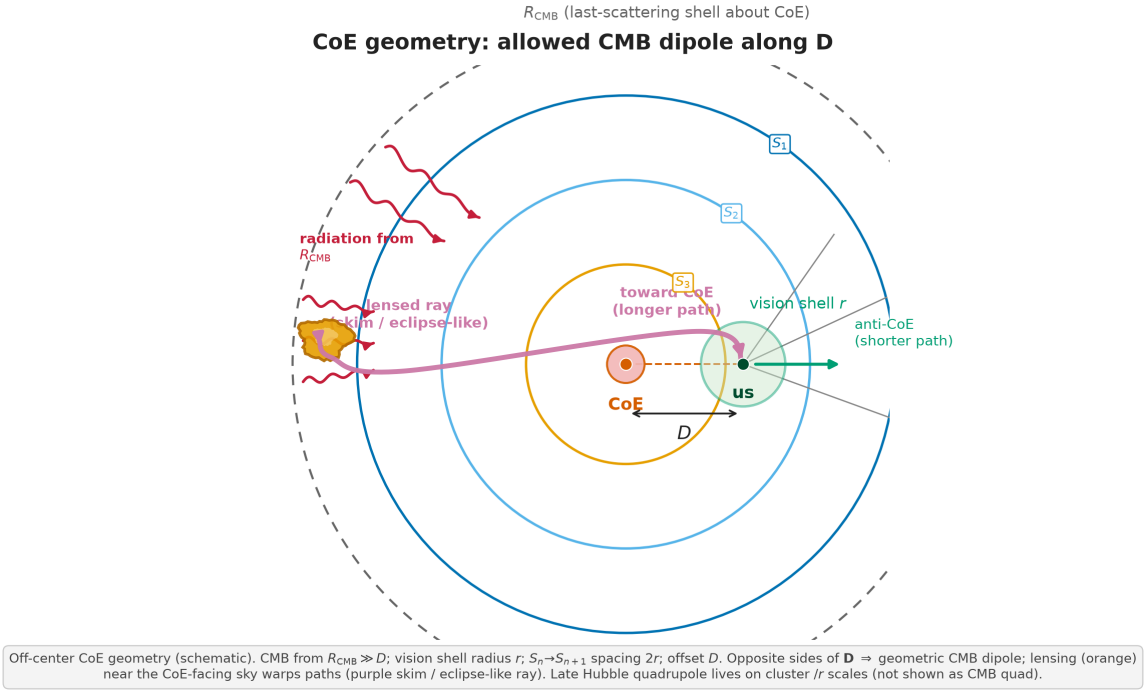


Figure 18: CMB/CoE shell schematic. Vision shell radius r; S_n to S_{n+1} spacing $2r$. CMB parked as open adversary.

FIGURE TO ADD (1/2)

Caption:

Figure 19

FIGURE TO ADD (2/2)

Caption:

Figure 20